1. *Design a Mod-12 synchronous counter using:*
   (i) JK flip-flops
   (ii) D flip-flops

- Construct and present the circuit diagrams for both designs.
- Illustrate the timing diagrams to show the counter operation.
- Perform a comparative analysis of both implementations in terms of design complexity, hardware requirements, and performance.
- Provide meaningful insights and comments based on your comparison.

2. *Design a circuit to generate the binary sequence: 1000111010001110l0… using JK flip-flops.* Implement the design using:
   (i) **Direct Logic**
   (ii) **Indirect Logic**
   Present the circuit diagrams for both approaches and explain the logic behind each design.
3. *Implement the sequence from Question 2 using a **Universal Shift Register**.* Design the circuit and justify your approach by explaining how the shift register facilitates the sequence generation.

4. *Design a synchronous sequential circuit using Mealy model to detect the sequence **0110**, considering both overlapping and non-overlapping cases.*

- Use **D flip-flops** in your design.
- Provide state diagrams, transition tables, and circuit diagrams for both cases.
- Discuss the differences in behavior and implementation between overlapping and non-overlapping detection.

5. *Re-design the synchronous sequential circuit from Question 4 using the **Moore model** to detect the sequence **0110**, considering both overlapping and non-overlapping cases.*
   - Use **D flip-flops** in your implementation.
   - Provide the **state diagram**, **state transition table**, and **circuit diagram** for each case.
   - Explain how the output is generated in the Moore model and compare its behavior with the Mealy model implementation.